An outcome surrogate for granular interactions: proper orthogonal decomposition with Gaussian process regression

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Abstract

We describe the construction of a parametric surrogate for the outcome of a granular interaction computed by a three-dimensional particle dynamics simulation. The particle state is interpolated onto a coarse Eulerian grid, the resulting field ensemble is reduced by proper orthogonal decomposition (POD), and the map from interaction parameters to modal coefficients is regressed by Gaussian process (GP) regression. The purpose of the document is to fix notation and dimensions; measured results from a pilot ensemble are reported in Section 6.

1 Problem statement and notation

Let $d = 3$ denote the spatial dimension and $\Omega \subset \mathbb{R}^d$ the simulation domain, a rectangular box. The full-order model (FOM) advances n_p particles subject to gravity, contact forces and boundary contacts. Particle $i \in \{1, \dots, n_p\}$ carries a position $\boldsymbol{x}_i \in \mathbb{R}^d$, a velocity $\boldsymbol{v}_i \in \mathbb{R}^d$ and a mass $m_i > 0$. Nothing below depends on the particular force law, and the formulation applies unchanged if the FOM is later extended with rigid bodies or continuum fields.

The state is stored with the degrees of freedom of each particle contiguous, position followed by velocity:

$$\boldsymbol{q}(t) \in \mathbb{R}^{n_q}, \quad n_q = 2dn_p, \qquad \boldsymbol{q} = \left[\, \boldsymbol{x}_1, \boldsymbol{v}_1,\ \boldsymbol{x}_2, \boldsymbol{v}_2,\ \dots,\ \boldsymbol{x}_{n_p}, \boldsymbol{v}_{n_p} \,\right], \tag{1}$$

so that, with components indexed from 1,

$$q_{2d(i-1)+j} = (\boldsymbol{x}_i)_j, \qquad q_{2d(i-1)+d+j} = (\boldsymbol{v}_i)_j, \qquad j = 1, \dots, d. \tag{2}$$

For $d = 3$ the stride between consecutive particles is $2d = 6$. This is the layout used by the implementation; it is recorded here because every index expression in Section 2 inherits it. For the ensemble reported below $n_p = 3375$, hence $n_q = 20\,250$.

Each realization of the FOM is determined by a parameter vector

$$\boldsymbol{\theta} \in \mathcal{P} \subset \mathbb{R}^p, \qquad p = 7, \tag{3}$$

where $\mathcal{P}$ is a product of intervals. The components of $\boldsymbol{\theta}$ are the $d = 3$ components of the initial velocity applied to every particle, the release height, the per-particle mass, the coefficient of restitution $e \in [0, 1]$, and the Coulomb friction coefficient $\mu \geq 0$. The first four are *actions*, in the sense that an agent acting in the environment would select them; the last three are *material* parameters, which system identification would instead infer from observation.

We seek a surrogate for a coarse-grained functional of the terminal state, not for $\boldsymbol{q}$ itself. This is deliberate: $\boldsymbol{q}$ is chaotic, whereas coarse-grained functionals of a granular assembly are reproducible.

2 Interpolation onto the grid

Let the grid have N_j nodes along axis j, covering Ω inclusively, so the spacing is $h_j = (\Omega_j^{\max} - \Omega_j^{\min})/(N_j - 1)$. Nodes are labelled by a single index

$$\alpha \in \{1, \ldots, N\}, \qquad N = N_1 N_2 N_3, \tag{4}$$

obtained from the triple $(\alpha_1, \alpha_2, \alpha_3)$ of per-axis indices by $\alpha = (\alpha_1 - 1)N_2N_3 + (\alpha_2 - 1)N_3 + \alpha_3$. Presently $(N_1, N_2, N_3) = (32, 32, 16)$ and $N = 16\,384$. Let $\boldsymbol{y}_\alpha \in \mathbb{R}^d$ denote the position of node α.

Define $n_c = 1 + d = 4$ *channels*: the mass density and the d components of the momentum density. Particle quantities are interpolated onto the grid with the product of one-dimensional hat functions,

$$w_{i\alpha} = \prod_{j=1}^{d} \max\left(0,\ 1 - \frac{|(\boldsymbol{x}_i)_j - (\boldsymbol{y}_\alpha)_j|}{h_j}\right) \geq 0, \tag{5}$$

which are nonzero on the $2^d = 8$ nodes surrounding $\boldsymbol{x}_i$ and form a partition of unity,

$$\sum_{\alpha=1}^{N} w_{i\alpha} = 1 \qquad \text{for every } i. \tag{6}$$

The interpolation is then

$$\rho_\alpha = \sum_{i=1}^{n_p} m_i w_{i\alpha}, \qquad (\rho u_j)_\alpha = \sum_{i=1}^{n_p} m_i (\boldsymbol{v}_i)_j\, w_{i\alpha}, \qquad j = 1, \ldots, d, \tag{7}$$

and (6) gives exact conservation of mass,

$$\sum_{\alpha=1}^{N} \rho_\alpha = \sum_{i=1}^{n_p} m_i \sum_{\alpha=1}^{N} w_{i\alpha} = \sum_{i=1}^{n_p} m_i. \tag{8}$$

Assigning each particle wholly to its nearest node also satisfies (8) but is discontinuous in $\boldsymbol{x}_i$; the weights (5) are continuous, which is required for the ensemble of Section 3 to depend continuously on $\boldsymbol{\theta}$.

Write $\boldsymbol{s}_c \in \mathbb{R}^N$ for the field of channel $c \in \{1, \ldots, n_c\}$, with $c = 1$ the mass channel and $c = 1 + j$ the jth momentum component, and $\boldsymbol{s} \in \mathbb{R}^{n_c N}$ for their concatenation, $\dim \boldsymbol{s} = 65\,536$. The map $\boldsymbol{q} \mapsto \boldsymbol{s}$ defined by (7) is denoted $\mathcal{E} : \mathbb{R}^{n_q} \to \mathbb{R}^{n_c N}$.

The momentum channels are retained because $\boldsymbol{s}_1$ alone is not a Markov state: two assemblies with identical mass fields and different velocity fields evolve differently. They are, however, informative only while the material is in motion; see Section 6.

3 Reduction: proper orthogonal decomposition

3.1 Snapshot ensemble

Let $\{\boldsymbol{\theta}^{(m)}\}_{m=1}^{M}$ be a design of experiments in $\mathcal{P}$, obtained here by a scrambled Sobol sequence, a quasi-Monte Carlo (QMC) construction whose marginal projections are uniform by construction. For each m the FOM is advanced to a terminal time T and interpolated by (7), giving snapshots $\boldsymbol{s}_c^{(m)} \in \mathbb{R}^N$.

For each channel c independently, collect the M snapshots as the rows of

$$X_c \in \mathbb{R}^{M\times N}, \qquad (X_c)_{m\alpha} = \left(\boldsymbol{s}_c^{(m)}\right)_\alpha, \tag{9}$$

that is, entry (m,α) of X_c is the value of channel c at grid node α in realization m: the row index runs over realizations, the column index over grid nodes. Subtract the ensemble mean field from every row,

$$\tilde{X}_c = X_c - \mathbf{1}_M \bar{\boldsymbol{s}}_c^\top, \qquad \bar{\boldsymbol{s}}_c = \frac{1}{M}\sum_{m=1}^{M} \boldsymbol{s}_c^{(m)} \in \mathbb{R}^N, \tag{10}$$

with $\mathbf{1}_M \in \mathbb{R}^M$ the vector of ones, so that $\mathbf{1}_M \bar{\boldsymbol{s}}_c^\top \in \mathbb{R}^{M\times N}$ repeats $\bar{\boldsymbol{s}}_c^\top$ in each of its M rows.

A separate basis is constructed for each channel. A single basis spanning all n_c channels jointly would require a relative scaling between mass density and momentum density. Such a scaling is not physically determined: $\rho \geq 0$ whereas ρu_j is sign-indefinite, and the two have different units and substantially different dynamic ranges. Under any fixed scaling the channel assigned the larger magnitude dominates the singular spectrum, and the fraction of captured variance ceases to be a meaningful diagnostic. Independent bases additionally permit channel-dependent truncation ranks.

3.2 Singular value decomposition

The singular value decomposition (SVD) of the centered snapshot matrix is

$$\tilde{X}_c = U_c \Sigma_c V_c^\top, \qquad U_c \in \mathbb{R}^{M\times r_c}, \quad \Sigma_c \in \mathbb{R}^{r_c\times r_c}, \quad V_c \in \mathbb{R}^{N\times r_c}, \tag{11}$$

with $U_c^\top U_c = V_c^\top V_c = I_{r_c}$, $\Sigma_c = \operatorname{diag}(\sigma_{c,1},\dots,\sigma_{c,r_c})$, $\sigma_{c,1} \geq \cdots \geq \sigma_{c,r_c} > 0$, and $r_c = \operatorname{rank}\tilde{X}_c \leq \min(M-1, N)$. The bound $M-1$ rather than M follows from (10), which removes one degree of freedom from the row space. For the present ensemble $M = 150 \ll N$, so $r_c \leq 149$ and the SVD is computed in economy form at cost $O(M^2 N)$.

The columns $\boldsymbol{\phi}_{c,k} \in \mathbb{R}^N$ of V_c are the POD modes of channel c: orthonormal spatial fields on the grid. Retaining the leading $K_c \leq r_c$ modes defines

$$\Phi_c = \left[\boldsymbol{\phi}_{c,1},\dots,\boldsymbol{\phi}_{c,K_c}\right] \in \mathbb{R}^{N\times K_c}, \qquad \Phi_c^\top \Phi_c = I_{K_c}, \tag{12}$$

and the modal coefficients of a snapshot follow by orthogonal projection,

$$\boldsymbol{a}_c^{(m)} = \Phi_c^\top \left(\boldsymbol{s}_c^{(m)} - \bar{\boldsymbol{s}}_c\right) \in \mathbb{R}^{K_c}, \qquad \boldsymbol{s}_c^{(m)} \approx \bar{\boldsymbol{s}}_c + \Phi_c \boldsymbol{a}_c^{(m)}. \tag{13}$$

The truncation rank K_c is chosen per channel; it is not assumed equal across c.

Two properties motivate this basis. First, by the Eckart–Young theorem the truncation (12) is optimal: among all rank-K_c matrices,

$$\min_{\operatorname{rank} Y \leq K_c} \left\| \tilde{X}_c - Y \right\|_F^2 = \sum_{k=K_c+1}^{r_c} \sigma_{c,k}^2, \tag{14}$$

where $\|\cdot\|_F$ is the Frobenius norm. No other K_c-dimensional linear subspace represents the ensemble with smaller squared error. Second, (14) makes the truncation error available from the spectrum alone, before any regression is attempted.

3.3 Retained variance, and the term "energy"

Because

$$\left\|\tilde{X}_c\right\|_F^2 = \sum_{k=1}^{r_c} \sigma_{c,k}^2 = \sum_{m=1}^{M} \left\|\boldsymbol{s}_c^{(m)} - \bar{\boldsymbol{s}}_c\right\|_2^2, \tag{15}$$

the natural truncation diagnostic is the fraction of that total carried by the retained modes,

$$\eta_c(K) = \frac{\sum_{k=1}^{K} \sigma_{c,k}^2}{\sum_{k=1}^{r_c} \sigma_{c,k}^2} \;=\; 1 - \frac{\min_{\operatorname{rank} Y \le K} \left\|\tilde{X}_c - Y\right\|_F^2}{\left\|\tilde{X}_c\right\|_F^2} \;\in [0,1]. \tag{16}$$

By (15) the denominator is $(M-1)$ times the sample variance of the field summed over all N nodes, so η_c is exactly the *fraction of ensemble variance retained*.

The quantity commonly reported as "retained energy", and quoted as a percentage, is the same $\eta_c(K)$ of (16); the two names refer to one quantity and there is no separate formula. The name is literal only in the setting where POD was first popularized, in which the snapshots are velocity fields: then $\sum_k \sigma_k^2$ is proportional to the total fluctuating kinetic energy of the ensemble, and truncation discards a definite amount of it. For a mass-density channel no energy is involved and the term is an analogy resting only on η_c being a normalized sum of squares. We therefore report η_c as retained variance, and note the synonym once, here.

η_c is a statement about the ensemble used to build the basis. It is not a generalization estimate, and Section 3.4 does not use it as one.

3.4 Error measures

All errors are measured between fields on the grid, in $\mathbb{R}^N$: the reference is the interpolated field $\boldsymbol{s}_c^{(m)}$ of (7), not the particle state $\boldsymbol{q}$. This is by construction rather than by convenience. The grid field is the declared quantity of interest, so the information discarded by $\mathcal{E}$ is part of the problem statement and not an error to be charged against the reduction or the regression. A metric comparing predictions against $\boldsymbol{q}$ would instead measure the coarse-graining, which is a modeling choice made in Section 1.

Let $\mathcal{T}$ be a set of realizations held out from the construction of $\bar{\boldsymbol{s}}_c$, of Φ_c, and of the regressor of Section 4. Two errors are reported separately. The first replaces the true coefficients into (13) and so isolates the truncation:

$$\varepsilon_c^{\mathrm{proj}}(K_c) = \left(\frac{\sum_{m\in\mathcal{T}} \left\| \left[\bar{\boldsymbol{s}}_c + \Phi_c\Phi_c^\top(\boldsymbol{s}_c^{(m)} - \bar{\boldsymbol{s}}_c)\right] - \boldsymbol{s}_c^{(m)} \right\|_2^2}{\sum_{m\in\mathcal{T}} \left\|\boldsymbol{s}_c^{(m)}\right\|_2^2} \right)^{1/2}, \tag{17}$$

in which the bracketed term is the orthogonal projection of the true field onto the affine subspace $\bar{\boldsymbol{s}}_c + \operatorname{range}\Phi_c$. The second uses the regressed coefficients $\boldsymbol{m}_c(\boldsymbol{\theta}^{(m)})$ of (21) and so measures the deployed surrogate:

$$\varepsilon_c^{\mathrm{sur}}(K_c) = \left(\frac{\sum_{m\in\mathcal{T}} \left\| \left[\bar{\boldsymbol{s}}_c + \Phi_c\boldsymbol{m}_c(\boldsymbol{\theta}^{(m)})\right] - \boldsymbol{s}_c^{(m)} \right\|_2^2}{\sum_{m\in\mathcal{T}} \left\|\boldsymbol{s}_c^{(m)}\right\|_2^2} \right)^{1/2} \;\ge\; \varepsilon_c^{\mathrm{proj}}(K_c), \tag{18}$$

the inequality holding because the projection is the minimizer over coefficients. The gap between (17) and (18) is attributable to the regression alone, so the two remedies – enlarging K_c and improving the regressor – can be prioritized from measurement rather than assumption.

Three choices in (17)–(18) are deliberate. The norms are taken on the *uncentered* field, because that is what the surrogate must produce; reporting the error of $\boldsymbol{s}_c - \bar{\boldsymbol{s}}_c$ instead would credit the method with the mean field, which is free. The ratio is formed once over the whole of $\mathcal{T}$ rather than by averaging per-realization ratios, so that realizations of small total mass, whose individual ratios are ill-conditioned, do not dominate. And $\mathcal{T}$ is excluded from $\bar{\boldsymbol{s}}_c$ and Φ_c: computing either from all M realizations leaks the test set into the basis and understates (17).

4 Regression: Gaussian process on modal coefficients

Composing (13) with the parameter-to-state map reduces the task to the regression of K_c scalar functions of $\boldsymbol{\theta}$,

$$f_{c,k} : \mathcal{P} \subset \mathbb{R}^p \to \mathbb{R}, \qquad f_{c,k}\big(\boldsymbol{\theta}^{(m)}\big) = \big(\boldsymbol{a}_c^{(m)}\big)_k, \qquad k = 1, \dots, K_c. \tag{19}$$

Each $f_{c,k}$ is modeled as an independent GP: a prior over functions under which any finite collection of function values is jointly Gaussian, specified by a mean function, taken zero after centering, and a covariance (kernel) function $\kappa : \mathcal{P} \times \mathcal{P} \to \mathbb{R}$.

We adopt the Matérn kernel with smoothness parameter $\nu = 5/2$ and automatic relevance determination (ARD), that is, one length scale per input,

$$\kappa(\boldsymbol{\theta}, \boldsymbol{\theta}') = \sigma_f^2 \left(1 + \sqrt{5}\,\tau + \tfrac{5}{3}\tau^2\right) e^{-\sqrt{5}\,\tau}, \qquad \tau = \left(\sum_{i=1}^{p} \frac{(\theta_i - \theta_i')^2}{\ell_i^2}\right)^{1/2}, \tag{20}$$

with signal variance $\sigma_f^2 > 0$ and length scales $\ell_i > 0$. Sample paths of (20) are twice differentiable, a weaker and, for a system with contact, more defensible assumption than the analyticity implied by a squared-exponential kernel. Observations are modeled as $y^{(m)} = f_{c,k}(\boldsymbol{\theta}^{(m)}) + \epsilon^{(m)}$ with $\epsilon^{(m)} \sim \mathcal{N}(0, \sigma_n^2)$ independent. Here σ_n^2 absorbs the irreducible scatter of a chaotic system and is therefore a physically meaningful quantity rather than a regularization device.

Let $\Theta \in \mathbb{R}^{M_{\rm tr} \times p}$ collect the $M_{\rm tr}$ training parameter vectors, $\boldsymbol{y} \in \mathbb{R}^{M_{\rm tr}}$ the corresponding coefficients for a fixed pair (c, k), $\mathrm{K} \in \mathbb{R}^{M_{\rm tr} \times M_{\rm tr}}$ the Gram matrix $\mathrm{K}_{mn} = \kappa(\boldsymbol{\theta}^{(m)}, \boldsymbol{\theta}^{(n)})$, and $\boldsymbol{\kappa}_* \in \mathbb{R}^{M_{\rm tr}}$ the vector $(\boldsymbol{\kappa}_*)_m = \kappa(\boldsymbol{\theta}^{(m)}, \boldsymbol{\theta}_*)$ at a query $\boldsymbol{\theta}_*$. The posterior is Gaussian with mean and variance

$$m_{c,k}(\boldsymbol{\theta}_*) = \boldsymbol{\kappa}_*^\top \left(\mathrm{K} + \sigma_n^2 I_{M_{\rm tr}}\right)^{-1} \boldsymbol{y}, \tag{21}$$

$$v_{c,k}(\boldsymbol{\theta}_*) = \kappa(\boldsymbol{\theta}_*, \boldsymbol{\theta}_*) - \boldsymbol{\kappa}_*^\top \left(\mathrm{K} + \sigma_n^2 I_{M_{\rm tr}}\right)^{-1} \boldsymbol{\kappa}_*. \tag{22}$$

Hyperparameters $(\sigma_f, \ell_1, \dots, \ell_p, \sigma_n)$ are obtained by maximizing the log marginal likelihood

$$\log p(\boldsymbol{y} \mid \Theta) = -\tfrac{1}{2}\boldsymbol{y}^\top A^{-1} \boldsymbol{y} - \tfrac{1}{2}\log\det A - \tfrac{M_{\rm tr}}{2}\log 2\pi, \qquad A = \mathrm{K} + \sigma_n^2 I, \tag{23}$$

at cost $O(M_{\rm tr}^3)$ per evaluation through the Cholesky factorization of A. For $M_{\rm tr} = O(10^2\text{–}10^3)$ this is negligible.

Three properties are relevant here. First, (22) supplies a predictive variance in closed form, which permits the model's stated uncertainty to be compared against the measured scatter of the FOM. Second, exact inference is available at the sample sizes affordable, where a high-capacity parametric regressor would be data-starved. Third, the fitted length scales rank the sensitivity of the outcome to each component of $\boldsymbol{\theta}$, since $\ell_i \to \infty$ removes input i from (20); this follows from (23) without additional sampling.

Symbol	Meaning	Value
d	spatial dimension	3
n_p	particles per realization	3375
n_q	FOM state dimension, $2dn_p$	20 250
p	parameters, $\dim \boldsymbol{\theta}$	7
n_c	channels, $1+d$	4
(N_1, N_2, N_3)	grid nodes per axis	$(32, 32, 16)$
N	nodes per channel, $N_1N_2N_3$	16 384
$\dim \boldsymbol{s}$	interpolated state, n_cN	65 536
M	realizations (snapshots)	150
N_t	recorded instants per realization	10
r_c	$\operatorname{rank} \tilde{X}_c \le \min(M-1, N)$	≤ 149
K_c	retained modes, channel c	20
n_{GP}	scalar GPs, $\sum_{c\in\mathcal{C}} K_c$	20

Table 1: Dimensions of the full-order model, the grid interpolation, the reduced basis and the regressor.

5 The composed surrogate and its dimensions

The surrogate for channel c is

$$\hat{\boldsymbol{s}}_c(\boldsymbol{\theta}) = \bar{\boldsymbol{s}}_c + \Phi_c \boldsymbol{m}_c(\boldsymbol{\theta}), \qquad \boldsymbol{m}_c(\boldsymbol{\theta}) = \left[m_{c,1}(\boldsymbol{\theta}), \dots, m_{c,K_c}(\boldsymbol{\theta})\right]^\top \in \mathbb{R}^{K_c}. \tag{24}$$

Because (24) is affine in the coefficients, $\hat{\boldsymbol{s}}_c$ is the exact conditional mean of the reduced representation, $\mathbb{E}[\boldsymbol{s}_c \mid \boldsymbol{\theta}] = \bar{\boldsymbol{s}}_c + \Phi_c \mathbb{E}[\boldsymbol{a}_c \mid \boldsymbol{\theta}]$; the independence assumed across k therefore does *not* bias the predicted mean field. It affects only the predictive covariance, which under independence is the rank-K_c matrix

$$\operatorname{Cov}[\boldsymbol{s}_c \mid \boldsymbol{\theta}] \approx \Phi_c \operatorname{diag}(v_{c,1}(\boldsymbol{\theta}), \dots, v_{c,K_c}(\boldsymbol{\theta}))\, \Phi_c^\top \in \mathbb{R}^{N\times N}, \tag{25}$$

and which omits any variance in the discarded subspace.

The number of scalar GPs required is

$$n_{\mathrm{GP}} = \sum_{c\in\mathcal{C}} K_c, \tag{26}$$

where $\mathcal{C} \subseteq \{1, \dots, n_c\}$ is the set of predicted channels. Predicting all channels at a common rank K gives $n_{\mathrm{GP}} = n_cK = 4K$, and the surrogate is a map $\mathbb{R}^p \to \mathbb{R}^{4K}$ followed by n_c independent affine liftings $\mathbb{R}^K \to \mathbb{R}^N$. For the terminal-state task of Section 6 we take $\mathcal{C} = \{1\}$ and $K_1 = 20$, so $n_{\mathrm{GP}} = 20$ and the surrogate is $\mathbb{R}^7 \to \mathbb{R}^{20} \to \mathbb{R}^{16\,384}$. Table 1 collects the dimensions.

6 Measured results

6.1 Spectra

For $M = 150$ realizations, a basis built from $M_{\mathrm{tr}} = 120$ and evaluated on the $|\mathcal{T}| = 30$ held out, (17) gives for the mass channel $c = 1$ at terminal time $T = 2\,\mathrm{s}$:

K_1	0 (mean field only)	5	10	20	40
$\varepsilon_1^{\mathrm{proj}}$	0.871	0.516	0.381	0.243	0.192

with $\eta_1 = 0.90$ at $K_1 = 11$ and $\eta_1 = 0.99$ at $K_1 = 51$. A linear basis is adequate for this channel: $K_1 = 20$ reduces a 16 384-dimensional field to a relative error of 0.24, against 0.87 for the mean field alone.

The momentum channels admit no useful representation at T, with $\varepsilon_c^{\mathrm{proj}}(20) \approx 0.98$ for $c = 2, 3, 4$. This reflects the state and not the method. The ratio of root-mean-square (RMS) momentum density to RMS mass density falls from 2.29 at $t = 0.2\,\mathrm{s}$ to 0.018 at $t = 2\,\mathrm{s}$: the assembly is at rest and $\rho\boldsymbol{u}$ is residual creep. Evaluated at $t = 0.2\,\mathrm{s}$ the same construction gives $\varepsilon_c^{\mathrm{proj}}(20) = 0.35$–$0.40$ for $c = 2, 3, 4$ against 0.34 for $c = 1$. The momentum channels are thus informative during transport and vacuous at rest, consistent with their role in Section 2. Since the initial velocity enters $\boldsymbol{\theta}$ directly, no information is lost by taking $\mathcal{C} = \{1\}$ for the terminal-state surrogate.

6.2 Composed surrogate

With $\mathcal{C} = \{1\}$, $K_1 = 20$, $n_{\mathrm{GP}} = 20$, $M_{\mathrm{tr}} = 120$ and $|\mathcal{T}| = 30$:

Quantity	Relative error
Mean field only, $K_1 = 0$	0.871
$\varepsilon_1^{\mathrm{proj}}(20)$, truncation only, Eq. (17)	0.243
$\varepsilon_1^{\mathrm{sur}}(20)$, composed surrogate, Eq. (18)	0.381
Regression alone, in coefficient space	0.350

and $\eta_1(20) = 0.968$. The truncation and regression contributions are comparable, so neither refining the basis nor improving the regressor alone would reduce $\varepsilon_1^{\mathrm{sur}}$ substantially. The per-mode coefficient of determination on $\mathcal{T}$ is $R^2 \geq 0.93$ for $k \leq 5$, decaying to $R^2 \approx 0.73$–0.86 by $k = 9$: the leading modes carry smooth, parameter-driven structure and the tail carries realization-specific detail.

Pooling all modes, the standardized residuals $z_{c,k} = (m_{c,k} - a_{c,k})/v_{c,k}^{1/2}$ satisfy $\Pr(|z| \leq 1) = 0.743$ and $\Pr(|z| \leq 2) = 0.918$ against nominal 0.683 and 0.954, with $\mathrm{RMS}(z) = 1.16$. The predictive variance (22) is close to calibrated and mildly overconfident.

The leading modes admit direct physical interpretation. Over the training set the correlation between $(\boldsymbol{a}_1)_k$ and the components of $\boldsymbol{\theta}$ is maximal at 0.673 for $k = 1$ against the first velocity component, 0.776 for $k = 2$ against the second, and 0.898 for $k = 6$ against the particle mass; the singular values $\sigma_{1,1} = 107.8$ and $\sigma_{1,2} = 101.7$ are nearly equal, as required by the symmetry of Ω and $\mathcal{P}$ under exchange of the first two coordinates.

Ranking inputs by ARD relevance, the σ^2-weighted mean of $1/\ell_i$ over retained modes,

$$v_x\,(1.94) \;\approx\; v_y\,(1.66) \;>\; \mu\,(1.25) \;>\; v_z\,(0.90) \;>\; m\,(0.78) \;\approx\; e\,(0.74) \;\gg\; z_0\,(0.23). \tag{27}$$

Aggregating ℓ_i rather than $1/\ell_i$ is incorrect: the basis assigns the two horizontal velocity components to separate modes of nearly equal energy, so a weighted mean of length scales ranks v_x above v_y by the margin $\sigma_{1,1}^2 - \sigma_{1,2}^2$ alone, an artifact of the aggregation rather than a property of the system.

6.3 Irreducible scatter

The errors above are uninterpretable without the scatter of the FOM itself at fixed $\boldsymbol{\theta}$. We generate an ensemble of E realizations sharing one $\boldsymbol{\theta}$ and differing only by an independent random displacement of magnitude ϵ applied to every initial particle position, and measure separation in two senses: the root-mean-square (RMS) separation of corresponding particles,

$$D_{\text{grain}}(t) = \left(\frac{1}{n_p} \sum_{i=1}^{n_p} \left\| \boldsymbol{x}_i^{(e)}(t) - \boldsymbol{x}_i^{(0)}(t) \right\|_2^2 \right)^{1/2}, \tag{28}$$

and the relative difference of the mass field, $D_{\text{field}}(t) = \left\| \boldsymbol{s}_1^{(e)}(t) - \boldsymbol{s}_1^{(0)}(t) \right\|_2 / \left\| \boldsymbol{s}_1^{(0)}(t) \right\|_2$.

For $\epsilon = 10^{-3} r$, with r the particle radius, D_{grain} amplifies by a factor 3.8×10^3, from 10^{-5} m to 3.81×10^{-2} m, the latter being 1.9 particle diameters. The growth is not a single exponential: the local rate $\mathrm{d}(\ln D_{\text{grain}})/\mathrm{d}t$ is $23.3\,\mathrm{s}^{-1}$ on $t \in [0.02, 0.06]$ s, falls to a sustained 5.6–6.4 s^{-1} on $[0.14, 0.60]$ s, corresponding to an e-folding time $\lambda^{-1} = 0.16$–0.18 s, and collapses to $0.17\,\mathrm{s}^{-1}$ thereafter. The plateau is reached, to within ten percent, at $t = 0.62$ s, which we take as the horizon over which individual positions remain predictable.

The plateau is set by jamming rather than by the assembly scale: growth ceases when the pile comes to rest and frictional contacts lock the packing, which is why D_{grain} saturates at 1.9 diameters and not at the extent of the domain. The estimate $\lambda^{-1} \ln(D_{\text{sat}}/\epsilon) = 1.38$ s consequently overstates the measured horizon, the early growth being faster than the sustained rate.

Over the same interval D_{field} reaches only 0.086 and the displacement of the field center of mass 0.009 grid nodes. Chaos destroys the trajectory and leaves the coarse-grained functional intact.

For the surrogate the relevant quantity is the scatter of realizations about their conditional mean, not the separation of two realizations. If $\boldsymbol{s}_1 = \boldsymbol{\mu} + \boldsymbol{n}$ with $\mathbb{E}[\boldsymbol{n}] = \mathbf{0}$ and $\boldsymbol{n}$ independent between realizations, then

$$\mathbb{E} \left\| \boldsymbol{s}_1^{(a)} - \boldsymbol{s}_1^{(b)} \right\|_2^2 = 2\, \mathbb{E}\, \|\boldsymbol{n}\|_2^2 , \tag{29}$$

so a model predicting $\boldsymbol{\mu}$ incurs $\|\boldsymbol{n}\|$ while the pairwise difference is larger by $\sqrt{2}$. With $\epsilon = 0.25r$, comparable to the lattice disorder, and $E = 16$, the measured pairwise difference is 0.245 and the measured RMS deviation about the ensemble mean is

$$\varepsilon_1^{\text{floor}} = \left(\frac{\sum_e \left\| \boldsymbol{s}_1^{(e)} - \bar{\boldsymbol{s}}_1^{\text{ens}} \right\|_2^2}{\sum_e \left\| \boldsymbol{s}_1^{(e)} \right\|_2^2} \right)^{1/2} = 0.168, \tag{30}$$

their ratio being 1.46 against the $\sqrt{2} = 1.414$ predicted by (29).

Relative to (30), $\varepsilon_1^{\text{proj}}(20) = 0.243$ is 1.45 times the floor and $\varepsilon_1^{\text{sur}}(20) = 0.381$ is 2.27 times it. Both exceed the floor, so both admit improvement; substituting the pairwise 0.245 for (30) would have indicated, incorrectly, that the surrogate was near-optimal. Since $\varepsilon_1^{\text{proj}}(40) = 0.192$ remains above 0.168, the truncation rank is not yet sufficient at $K_1 = 20$, and the gap $\varepsilon_1^{\text{sur}} - \varepsilon_1^{\text{proj}}$ indicates the regression as the larger of the two deficits at $M_{\text{tr}} = 120$.

7 Latent dynamics

7.1 Formulation

Section 4 regresses the terminal state directly from $\boldsymbol{\theta}$. An alternative is to learn a one-step map in the reduced coordinates and apply it repeatedly. Partition $\boldsymbol{\theta}$ as $\boldsymbol{\theta} = (\boldsymbol{\theta}_a, \boldsymbol{\theta}_m)$, with $\boldsymbol{\theta}_a \in \mathbb{R}^4$ the action components (the three initial velocity components and the release height) and $\boldsymbol{\theta}_m \in \mathbb{R}^3$ the material components (particle mass, restitution, friction). Let

$$\boldsymbol{z}(t) = \big[\, \boldsymbol{a}_1(t)^\top, \dots, \boldsymbol{a}_{n_c}(t)^\top \,\big]^\top \in \mathbb{R}^{n_z}, \qquad n_z = \sum_{c=1}^{n_c} K_c, \tag{31}$$

be the concatenated modal coordinates of all n_c channels, each obtained by (13). We regress

$$\mathcal{G} : \mathbb{R}^{n_z+3} \to \mathbb{R}^{n_z}, \qquad \boldsymbol{z}(t+\Delta) = \mathcal{G}\big(\boldsymbol{z}(t), \boldsymbol{\theta}_m\big), \tag{32}$$

component-wise by n_z independent Gaussian processes as in Section 4, and define the j-step prediction by composition,

$$\hat{\boldsymbol{z}}(t_0 + j\Delta) = \mathcal{G}^{\,j}\big(\boldsymbol{z}(t_0), \boldsymbol{\theta}_m\big), \tag{33}$$

with $\boldsymbol{z}(t_0)$ the projection of an observed initial field. Here $\Delta = 0.1\,\mathrm{s}$ and the solver timestep is $10^{-4}\,\mathrm{s}$, so each application of $\mathcal{G}$ advances 10^3 solver steps.

Two choices are consequential. First, all $n_c = 4$ channels are carried in (31), whereas Section 6 predicts only $c = 1$: the mass field alone is not a Markov state, since two assemblies with equal mass fields and unequal velocity fields evolve differently. Second, $\boldsymbol{\theta}_a$ is deliberately *withheld* from (32). Supplying it would permit $\mathcal{G}$ to ignore $\boldsymbol{z}$ and internally represent $\boldsymbol{\theta} \mapsto \boldsymbol{z}(t+\Delta)$, collapsing the construction onto the direct regression of Section 4 and rendering the sufficiency of $\boldsymbol{z}$ untestable.

7.2 Error decomposition

The error of (33) separates into three terms, each independently measurable, and the identity of the dominant term determines which remedy is appropriate. Writing $\varepsilon^{\text{proj}}$ for (17) evaluated at time t, $\varepsilon^{\text{step}}$ for the error of a single application of $\mathcal{G}$ to the *projected* state (teacher forcing), and $\varepsilon^{\text{roll}}$ for that of (33),

$$\underbrace{\varepsilon^{\text{proj}}}_{\text{truncation}} \;\leq\; \underbrace{\varepsilon^{\text{step}}}_{+\text{ regression}} \;\leq\; \underbrace{\varepsilon^{\text{roll}}}_{+\text{ accumulation}} . \tag{34}$$

At $K_1 = 16$ we measured $\varepsilon^{\text{proj}} = 0.4012$ and $\varepsilon^{\text{step}} = 0.4013$ at $t = 1\,\mathrm{s}$: the regression term is 10^{-4}, so the truncation term accounts for 86% of $\varepsilon^{\text{roll}} = 0.464$ and no quantity of additional training data could have improved the result. Raising the rank to $K_1 = 96$ reduced $\varepsilon^{\text{proj}}$ to 0.173 and left accumulation dominant, at which point reducing the number of compositions in (33) and increasing the density of the training design became the appropriate remedies.

7.3 Results

With $K_1 = 96$, $K_{2,3,4} = 24$ so that $n_z = 168$, $\Delta = 0.1\,\mathrm{s}$, $M = 1536$ realizations of $N_t = 10$ instants, and $M_{\text{tr}} = 900$ transitions drawn from the 12 744 available:

Quantity at $t = 1\,\mathrm{s}$	$\Delta = 0.05\,\mathrm{s}$, $M = 256$	$\Delta = 0.10\,\mathrm{s}$, $M = 1536$
$\varepsilon^{\text{proj}}$ (truncation)	0.173	0.108
$\varepsilon^{\text{step}}$ (teacher forced)	0.177	0.109
$\varepsilon^{\text{roll}}$ (rollout)	0.445	**0.174**
persistence, $\boldsymbol{z}(t+\Delta) = \boldsymbol{z}(t)$	1.762	1.531
accumulation, $\varepsilon^{\text{roll}} - \varepsilon^{\text{step}}$	0.268	0.065
$\varepsilon^{\text{floor}}$ for that configuration	0.168	0.154
$\varepsilon^{\text{roll}}/\varepsilon^{\text{floor}}$	2.65	**1.13**

$\varepsilon^{\text{roll}}$ is non-monotonic in t, attaining 0.33 at $t \approx 0.35\,\text{s}$ before decreasing: the impact and spreading phase is the least predictable part of the trajectory, after which the assembly jams and the state becomes easier to predict.

The improvement was obtained by halving the number of compositions in (33) and by contracting $\mathcal{P}$ by approximately 40% along each axis. The latter reduces $\varepsilon^{\text{step}}$ at fixed M_{tr}, since a fixed training set covers a smaller domain more densely, and simultaneously reduces $\varepsilon^{\text{proj}}$, since a less diverse ensemble is spanned by fewer modes ($\eta_1(96)$ rising from 0.9943 to 0.9974). It is purchased at the cost of generality: the contracted $\mathcal{P}$ has approximately 7% of the original volume in $p = 7$ dimensions, and $\varepsilon^{\text{roll}} = 0.174$ is meaningful only on the contracted domain.

We note that exact Gaussian process inference is $O(M_{\text{tr}}^3)$ per evaluation of (23), so M_{tr} cannot be raised in proportion to M. A larger M improves the basis and permits a better-covering subset, but reducing $\varepsilon^{\text{step}}$ at fixed M_{tr} requires contracting $\mathcal{P}$ rather than enlarging the design.

8 The irreducible floor

Every error reported above is quoted against $\varepsilon^{\text{floor}}$, and the definition of that quantity, its justification, and the manner of its measurement are given here in full, since a surrogate error is not interpretable without it.

8.1 Definition

Let $\boldsymbol{\theta}$ be fixed and let ω denote the microstate, that is, the particular initial packing of the assembly. The vector $\boldsymbol{\theta}$ specifies the *macrostate*; it does not determine ω, and many ω are consistent with one $\boldsymbol{\theta}$. The terminal field is therefore a random variable $\boldsymbol{s}_1(\boldsymbol{\theta}, \omega)$, and any surrogate $\hat{\boldsymbol{s}}_1(\boldsymbol{\theta})$ is a function of $\boldsymbol{\theta}$ alone. Write

$$\boldsymbol{\mu}(\boldsymbol{\theta}) = \mathbb{E}_\omega\big[\boldsymbol{s}_1(\boldsymbol{\theta}, \omega)\big], \qquad \boldsymbol{n}(\boldsymbol{\theta}, \omega) = \boldsymbol{s}_1(\boldsymbol{\theta}, \omega) - \boldsymbol{\mu}(\boldsymbol{\theta}), \qquad \mathbb{E}_\omega[\boldsymbol{n}] = \boldsymbol{0}. \tag{35}$$

8.2 Justification

Expanding the expected squared error of an arbitrary $\hat{\boldsymbol{s}}_1$ and using $\mathbb{E}_\omega[\boldsymbol{n}] = \boldsymbol{0}$ to annihilate the cross term,

$$\mathbb{E}_\omega \left\|\hat{\boldsymbol{s}}_1(\boldsymbol{\theta}) - \boldsymbol{s}_1(\boldsymbol{\theta}, \omega)\right\|_2^2 = \underbrace{\left\|\hat{\boldsymbol{s}}_1(\boldsymbol{\theta}) - \boldsymbol{\mu}(\boldsymbol{\theta})\right\|_2^2}_{\text{model error}} + \underbrace{\mathbb{E}_\omega \left\|\boldsymbol{n}(\boldsymbol{\theta}, \omega)\right\|_2^2}_{\text{irreducible}}. \tag{36}$$

The second term does not contain $\hat{\boldsymbol{s}}_1$. It is therefore unaffected by any change to the surrogate, and the first term is minimised, at zero, by $\hat{\boldsymbol{s}}_1 = \boldsymbol{\mu}$. The best attainable error is thus $\left(\mathbb{E}_\omega \|\boldsymbol{n}\|_2^2\right)^{1/2}$, and it is unattainable to improve upon because the information distinguishing realizations is absent from the input, not merely unexploited. In the terminology of uncertainty quantification this is the *aleatoric* component, as against the *epistemic* component represented by the first term.

8.3 Measurement

We generate E realizations sharing $\boldsymbol{\theta}$ and differing only by an independent random displacement of magnitude ϵ applied to each initial particle position, take ϵ comparable to the disorder of the packing itself so that the realizations are genuinely distinct members of one macrostate, and evaluate the normalized scatter about their sample mean,

$$\varepsilon^{\text{floor}} = \left(\frac{\sum_{e=1}^{E} \left\| \boldsymbol{s}_1^{(e)} - \bar{\boldsymbol{s}}_1^{\text{ens}} \right\|_2^2}{\sum_{e=1}^{E} \left\| \boldsymbol{s}_1^{(e)} \right\|_2^2} \right)^{1/2}, \qquad \bar{\boldsymbol{s}}_1^{\text{ens}} = \frac{1}{E} \sum_{e=1}^{E} \boldsymbol{s}_1^{(e)}. \tag{37}$$

Three properties of (37) require comment.

It is not the pairwise separation. Substituting the pairwise difference between two realizations overstates the floor by $\sqrt{2}$: from (35) with $\boldsymbol{n}$ independent between realizations,

$$\mathbb{E} \left\| \boldsymbol{s}_1^{(a)} - \boldsymbol{s}_1^{(b)} \right\|_2^2 = \mathbb{E} \left\| \boldsymbol{n}^{(a)} - \boldsymbol{n}^{(b)} \right\|_2^2 = 2\, \mathbb{E} \left\| \boldsymbol{n} \right\|_2^2, \tag{38}$$

whereas a model predicting $\boldsymbol{\mu}$ incurs $\mathbb{E}\|\boldsymbol{n}\|_2^2$. Measured on the contracted configuration at $t = 1\,\text{s}$ with $E = 16$: (37) gives 0.154 while the pairwise value is 0.225, a ratio of 1.463 against the $\sqrt{2} = 1.414$ predicted by (38). Quoting the pairwise figure would understate $\varepsilon^{\text{roll}}/\varepsilon^{\text{floor}}$ from 1.13 to 0.77, that is, would report a surrogate as having surpassed a bound it had not reached.

It is configuration-specific. $\varepsilon^{\text{floor}}$ depends on $\boldsymbol{\theta}$, on the horizon, and on the grid; it is 0.168 for the wide-$\mathcal{P}$ configuration at $t = 2\,\text{s}$ and 0.154 for the contracted configuration at $t = 1\,\text{s}$. It must be re-measured whenever any of these changes.

The sample mean biases it low. Using $\bar{\boldsymbol{s}}_1^{\text{ens}}$ in place of the unknown $\boldsymbol{\mu}$ underestimates the scatter by the factor $\sqrt{(E-1)/E}$, which is 0.968 at $E = 16$. The reported figures are not corrected for this, and it is small relative to the variation of $\varepsilon^{\text{floor}}$ with $\boldsymbol{\theta}$, which has not been characterized: all values here are measured at a single, central $\boldsymbol{\theta}$.

Finally, the floor is not a reconstruction bound. $\varepsilon^{\text{proj}} = 0.108$ lies *below* $\varepsilon^{\text{floor}} = 0.154$, and this is not contradictory: the former is the error of compressing a field that has already been realized, the latter the error of predicting which field will be realized. Only quantities of the second kind are bounded by (37).

9 Decoding to particles

9.1 The decode is a sampler

The interpolation $\mathcal{E}$ of Section 2 is many-to-one: distinct particle configurations produce identical fields, since (7) retains only $n_c N$ numbers of the n_q that describe $\boldsymbol{q}$, and here $n_c N / n_q = 3.2$ but the map is not injective because positions within a cell are not resolved. The preimage $\mathcal{E}^{-1}(\boldsymbol{s})$ is therefore a set, and no inverse exists. We construct instead a sampler

$$\mathcal{D} : \mathbb{R}^{n_c N} \to \mathbb{R}^{n_q}, \qquad \mathcal{D}(\boldsymbol{s}) \sim \pi\big(\cdot \mid \boldsymbol{s}\big), \tag{39}$$

drawing one element of that set. Consequently $\mathcal{D}(\mathcal{E}(\boldsymbol{q})) \neq \boldsymbol{q}$: the construction reproduces the field, not the configuration.

9.2 Construction

Given a predicted mass field $\boldsymbol{\rho}$ and momentum fields $(\boldsymbol{\rho u})_j$, and given that the per-particle mass m and hence the total mass $M = n_p m$ are known from $\boldsymbol{\theta}$:

(i) Non-negativity. A truncated expansion $\boldsymbol{\rho} = \bar{\boldsymbol{s}}_1 + \Phi_1 \boldsymbol{a}_1$ is not constrained to be non-negative and is not, so set $\tilde{\rho}_\alpha = \max(\rho_\alpha, 0)$. The discarded fraction $\sum_\alpha \max(-\rho_\alpha, 0) / \sum_\alpha \max(\rho_\alpha, 0)$ is 0.04 to 0.10 in the cases reported below.

(ii) Thresholding. Truncation also produces small positive values far from the material. Set $\tilde{\rho}_\alpha \leftarrow 0$ wherever $\tilde{\rho}_\alpha < \tau \max_\beta \tilde{\rho}_\beta$, with $\tau = 0.02$. This step is not cosmetic. Individually negligible, those values are spread over all $N = 16\,384$ nodes and collectively attract a substantial number of particles into a halo absent from the truth; and because they sit where $\tilde{\rho} \approx 0$, the quotient in (40) assigns those particles very large velocities. Measured at $t = 0.7\,\mathrm{s}$, thresholding reduces the error in the sampled spatial standard deviation from $0.092\,\mathrm{m}$ to $0.006\,\mathrm{m}$ and the sampled kinetic energy from 141 to 7.6 against a true value of 3.6.

(iii) Mass normalisation. Set $\hat{\rho}_\alpha = \tilde{\rho}_\alpha M / \sum_\beta \tilde{\rho}_\beta$, so that the decode conserves mass exactly even though steps (i) and (ii) do not.

(iv) Positions. Two samplers are used. The first draws a node index $\alpha \sim \hat{\rho}_\alpha / M$ and then a position uniformly in that node's cell; it reproduces the nodal mass in expectation and ignores the packing constraint. The second is described in Section 9.3.

(v) Velocities. With $w_\alpha(\boldsymbol{x})$ the same weights (5) used to deposit,

$$\boldsymbol{v}(\boldsymbol{x}) = \frac{\sum_\alpha w_\alpha(\boldsymbol{x})\,(\boldsymbol{\rho u})_\alpha}{\sum_\alpha w_\alpha(\boldsymbol{x})\,\hat{\rho}_\alpha}, \tag{40}$$

evaluated at the sampled position and set to zero where the denominator vanishes. Using the deposit's own weights rather than the nodal value keeps $\boldsymbol{v}$ continuous across cell boundaries.

9.3 Minimum-separation sampling

The uniform sampler of (iv) produces a configuration in which particles overlap: node spacing is $h_1 = h_2 = 0.065\,\mathrm{m}$ against a diameter $2r = 0.02\,\mathrm{m}$, so roughly 30 particles share a node and uniform placement within it is Poisson-like rather than packed. Such a configuration is unusable as an initial condition, for the reason quantified in Section 9.4.

We therefore impose

$$\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2 \;\geq\; 2r \quad \forall\, i \neq j, \qquad \Omega_j^{\min} + r \leq (\boldsymbol{x}_i)_j \leq \Omega_j^{\max} - r, \tag{41}$$

the second condition ensuring a particle centre is a full radius clear of each wall. Candidates are proposed as in (iv) and accepted only if (41) holds, with a spatial hash of cell size $2r$ so that a candidate need be tested against the $3^d = 27$ neighbouring cells only. Bridson's algorithm is the usual construction for a Poisson-disk set but generates a spatially *uniform* intensity; here the intensity is prescribed by $\hat{\rho}$ and the disk radius is fixed by the physics, so the density is carried by the proposal distribution instead.

Candidates violating the second condition of (41) must be *rejected* and not projected onto the boundary. Projection collapses all such candidates onto a face — and since the assembly rests on $\Omega_3^{\min}$, a large fraction of proposals are of this kind — which destroys the separation just enforced; we measured 36% of pairs in violation of the first condition despite a correct acceptance test.

The construction cannot produce a density exceeding the maximum packing fraction. Where $\hat{\rho}$ demands one, the requested particles cannot be placed; this indicates that the predicted field is locally unphysical, and the deficit is reported rather than concealed. In the cases below no deficit occurred: all $n_p = 3375$ particles were placed.

Measured on the resulting configuration: $\min_{i \neq j} \|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2 / 2r = 1.0002$, median nearest-neighbour distance $1.157 \times 2r$, and 0% of pairs in contact, against 73% for the uniform sampler at a median of $0.699 \times 2r$.

9.4 Restart validation

Whether $\mathcal{D}(\boldsymbol{s})$ is admissible as an initial condition is decided by integrating from it. Both configurations were written in the solver's state layout and advanced; the total energy of Section 3 (kinetic, gravitational and contact-elastic) at $t = 0$ and thereafter is

Sampler	pairs in contact	$E(0)$	behaviour
uniform	73%	14 893.5	90% of E released within 0.01 s
minimum-separation	0%	1 270.8	monotone decay

The factor of 11.7 is elastic energy stored in overlapping contacts by (7)'s inverse being taken carelessly, and it is converted immediately into kinetic energy. From the minimum-separation configuration the energy decreases monotonically over the full second integrated, $1270.8 \to 78.3$, with no contact-list overflow: the state is admissible.

9.5 Limitations

Two are intrinsic to the channels retained.

Only the first velocity moment is stored. The within-node velocity distribution — the granular temperature — is discarded by (7), so (40) assigns every particle in a neighbourhood the same velocity and the sampled kinetic energy is biased. Retaining a second-moment channel $\rho u_i u_j$ would recover it.

Enforcing (41) costs fidelity in the low-order moments, because the sampler cannot place particles at a density above the packing limit and displaces them outward instead. At $t = 1\,\mathrm{s}$ the error in the centroid rises from 0.025 m to 0.057 m and in the spatial standard deviation from 0.011 m to 0.045 m, against a particle diameter of 0.02 m. The uniform sampler is therefore preferable for visualisation and for bulk statistics, and the minimum-separation sampler whenever the configuration is to be integrated.

Nomenclature

ARD	automatic relevance determination (one kernel length scale per input)
FOM	full-order model (the particle dynamics simulation)
GP	Gaussian process
POD	proper orthogonal decomposition
QMC	quasi-Monte Carlo
RMS	root mean square
SVD	singular value decomposition